

A Physics-Informed Image Registration Framework for Hyperelastic Material Parameter Identification

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Abstract

The mechanical characterization of soft tissues is essential for surgical simulation and diagnosis. The work of Gao and Desai (2010) established a two-stage pipeline combining digital image correlation (DIC) with Ogden model fitting to estimate material properties of porcine liver. However, their approach relies on a heuristic estimate of the zero-strain reference state and separates displacement measurement from parameter identification, potentially propagating errors. We present a unified physics-informed neural network (PINN) framework that performs image registration and material parameter identification simultaneously. The method integrates a differentiable image similarity loss (replacing DIC), hyperelastic equilibrium constraints, boundary conditions, and a multi-step force loss. Using synthetic tissue images generated from finite element simulations that replicate the Gao–Desai experiment, we demonstrate accurate recovery of the Ogden parameters (μ and α) with errors below 0.1%, while implicitly resolving the zero-strain ambiguity. This work provides a proof-of-concept for end-to-end material characterization directly from image sequences, addressing key limitations of classical pipelines.

1 Introduction

Understanding the mechanical behavior of soft biological tissues is fundamental to the development of realistic surgical simulators, diagnostic tools, and patient-specific treatment planning [3]. Among soft tissues, the liver presents particular challenges due to its extremely low stiffness, which renders it susceptible to gravity-induced deformation even before external loading is applied [4].

The seminal work of Gao and Desai [4] established a rigorous experimental protocol for characterizing porcine liver tissue *ex vivo*. Their methodology employed digital image correlation (DIC) to track the natural surface texture of liver specimens during uniaxial tension tests. A key observation was the presence of a “maximum stretching band” in the incremental strain field, which they used to estimate the zero-strain reference state—a critical parameter given that the tissue is pre-compressed by gravity prior to test initiation. Material parameters for the Ogden hyperelastic model [8] were subsequently obtained by fitting the nominal stress-stretch curves extracted from DIC data.

While groundbreaking, this two-stage approach suffers from several limitations. First, the DIC displacement extraction and subsequent curve fitting are performed sequentially, allowing errors from the first stage to propagate to parameter identification. Second, the zero-strain state is determined approximately based on a heuristic, introducing uncertainty that can bias material parameters. Third, DIC processing requires careful parameter tuning and is not differentiable, precluding its integration into end-to-end learning pipelines.

Recent advances in physics-informed neural networks (PINNs) have demonstrated the potential to embed physical laws directly into learning algorithms [10]. In the context of biomechanics, PINNs have been applied to image registration with linear and nonlinear elasticity constraints [5], as well as to the discovery of hyperelastic constitutive models from sparse experimental data [11, 6]. These approaches treat the displacement field as the output of a neural network and enforce equilibrium equations as soft constraints, enabling simultaneous learning of kinematics and material parameters.

In this work, we present a unified PINN-based framework for hyperelastic image registration and material parameter identification. Our contributions are:

- A differentiable image similarity loss that replaces classical DIC, enabling direct optimization of displacement fields from raw image intensities.
- Integration of the Ogden hyperelastic model as a physics regularizer, enforcing equilibrium of linear momentum in the reference configuration with explicit gravity body force.
- A multi-step force loss that leverages reaction force measurements at multiple loading stages to break degeneracy in parameter identification.
- Validation on synthetic tissue images generated from finite element simulations that replicate the experimental conditions of Gao and Desai [4], demonstrating accurate recovery of both displacement fields and material parameters.

The remainder of this paper is organized as follows. Section 2 provides a detailed commentary on the Gao–Desai pipeline, highlighting its strengths and weaknesses. Section 3 reviews related work in PINN-based material identification and biomechanics-informed registration. Section 4 describes the mathematical formulation and our proposed PINN methodology. Section 5 presents the experimental setup and preliminary results. Sections 6 and 7 offer discussion and conclusions.

2 The Gao–Desai Experimental Pipeline

2.1 Experimental Protocol and DIC Analysis

Gao and Desai [4] conducted uniaxial tension tests on porcine liver tissue samples (rectangular prisms, height ≈ 25 mm, width and thickness 15–17 mm). Samples were extracted from frozen livers, thawed, and glued between two plates. The bottom plate was fixed while the top plate was displaced upward at a constant rate (1.27 mm/s, nominal strain rate 0.05 s^{-1}). Images of the tissue surface were captured at 3 frames/s using a digital camera.

To measure full-field deformation, they employed a global digital image correlation technique. The natural texture of the liver—visible connective tissue septa defining hepatic lobule boundaries—served as the correlation pattern, eliminating the need for artificial speckles. A Lagrangian mesh was defined on the reference image and tracked through subsequent images, yielding continuous displacement fields. The incremental deformation gradient $\mathbf{F}^n$ between consecutive images was computed, and cumulative Lagrangian strain $\mathbf{E}^n$ with respect to the initial image was obtained.

2.2 Zero-Strain State Estimation via Maximum Stretching Band

A critical insight was that the tissue in the initial image was not stress-free due to gravity pre-compression. As the top plate moved upward, different regions passed from compression through zero strain into tension. By examining the *incremental* strain field ΔE_{yy} between consecutive images, they observed a band of maximum incremental strain that migrated downward with increasing load. They hypothesized that this band corresponded to the region where the tissue was least stiff—i.e., near the zero-strain state. Based on this, they estimated the zero-strain reference for three horizontal planes (upper, middle, lower) by selecting images where the maximum band passed through each plane. The stretch ratio λ was then computed from the cumulative strain E_{yy} relative to that estimated zero-strain image, and nominal stress was obtained from force measurements divided by the original cross-sectional area.

2.3 Strengths of the Gao–Desai Approach

The methodology pioneered by Gao and Desai possesses several notable strengths that have made it a benchmark in soft tissue characterization. Foremost among these is the use of digital image correlation as a non-contact measurement technique. By relying on the natural connective tissue texture of porcine liver, the authors avoided the need for artificial speckles, which can delaminate or alter the mechanical properties of very soft tissues. The global DIC implementation provided full-field deformation data, revealing the spatially non-uniform strain fields caused by gravity pre-compression and tissue heterogeneity—a level of detail unattainable with traditional clip-on extensometers or actuator displacement measurements. Moreover, the observation of a “maximum stretching band” in the incremental strain fields offered a physically motivated, albeit approximate, method for estimating the elusive zero-strain reference state. The subsequent fitting of the Ogden hyperelastic model to the extracted stress–stretch

curves yielded material parameters that have since been used as reference values in numerous subsequent studies, underscoring the impact and reproducibility of their work.

2.4 Weaknesses and Limitations

Despite these strengths, the Gao–Desai pipeline exhibits several inherent limitations. The two-stage nature of the process—first extracting displacements via DIC, then fitting material parameters—creates a sequential error propagation pathway. Any inaccuracies introduced during the DIC correlation (e.g., interpolation artefacts, subset mismatches, or noise) directly corrupt the strain data used for constitutive fitting, and these errors cannot be corrected in the subsequent stage. Furthermore, the estimation of the zero-strain state via the maximum stretching band, while ingenious, remains heuristic and approximate; the exact position of zero strain relative to the band is unknown and may depend on material properties and sample geometry, introducing an uncontrolled source of bias. The DIC algorithm itself is not differentiable and requires manual tuning of parameters such as subset size and step size, precluding its integration into end-to-end gradient-based optimization frameworks. Additionally, the reaction force measurements acquired during the experiment are used only at the final fitting stage, not during displacement extraction, meaning that valuable information that could constrain the material parameters more tightly is underutilised. Finally, while gravity is acknowledged as the cause of pre-compression, it is not explicitly incorporated into the mechanical model during parameter identification; instead, its effect is absorbed into the zero-strain heuristic, leaving the potential for systematic error. These limitations motivate the exploration of unified approaches that can simultaneously leverage image data, physical laws, and force measurements in a single, differentiable framework.

3 Related Work

3.1 Physics-Informed Neural Networks for Solid Mechanics

Raissi et al. [10] introduced physics-informed neural networks (PINNs), which embed governing partial differential equations into neural network loss functions, enabling solution of both forward and inverse problems with limited data. In solid mechanics, PINNs have been applied to identify constitutive parameters from full-field deformation data [11, 6], reconstruct internal structures and defects [12], and model finite-strain plasticity [7]. The key advantage lies in their ability to incorporate physical knowledge directly into the learning process, requiring fewer training data points than purely data-driven approaches while maintaining consistency with governing equations.

For hyperelastic materials specifically, Song and Jin [11] introduced a PINN-based framework for identifying constitutive parameters of complex hyperelastic solids under plane stress conditions. Their approach emphasizes training with multi-modal synthetic experimental data consisting of full-field deformation (simulating DIC measurements) and loading history. Using the Arruda-Boyce eight-chain model [1] as a prototype, they demonstrated accurate parameter identification (errors below 5%) for samples with intricate geometries containing multiple internal defects, even with experimental noise levels up to 5%. Moon et al. [6] extended this direction by developing a PINN-based framework capable of discovering constitutive models using only sparse measurement data—specifically displacement and reaction force measurements from a single material test.

3.2 Biomechanics-Informed Image Registration

The intersection of biomechanics and medical image registration has emerged as a fertile research area. Classical registration methods rely on smoothness penalties such as diffusion or bending energy [2], but these purely mathematical regularizers do not guarantee physical plausibility. Recent work has sought to incorporate biomechanical constraints directly into registration frameworks.

Min et al. [5] introduced a biomechanics-informed non-rigid medical image registration algorithm using PINNs, investigating both linear and nonlinear elasticity models. Their framework, validated on clinical prostate MR images, treats registration and material property identification as forward and inverse problems respectively under the PINN paradigm. A key finding is that nonlinear elasticity shows no statistical significance over linear models in computing point-wise displacement vectors, though benefits may be patient-specific. Crucially, they formulated and solved the inverse parameter estimation problem

under a joint optimization scheme, demonstrating that accurate solutions can be obtained by locating saddle points in the loss landscape.

Rabineau et al. [9] addressed the challenge of identifying spatially varying elastic moduli from experimental full-field displacement data. Their mixed PINN formulation represents displacement and stress fields with distinct neural networks, incorporates multiple global mechanical equilibria to exploit reaction forces, and employs Fourier feature embeddings to capture high-frequency components. Validation on DIC-based experimental displacements from a perforated plate loaded to failure successfully identified homogeneous elastic constants in the initial regime and reconstructed Young’s modulus fields revealing early damage localization.

3.3 Positioning of the Present Work

Our framework is motivated directly by the limitations of the Gao–Desai pipeline described in Section 2. In particular, the proposed method addresses several key shortcomings of the classical two-stage approach. First, instead of separating displacement measurement and material parameter identification, the PINN formulation performs both tasks simultaneously within a unified optimization framework, reducing the risk of error propagation. Second, the differentiable image similarity loss replaces the non-differentiable DIC correlation procedure, allowing the displacement field to be learned directly from image intensities. Third, reaction force measurements are incorporated directly into training through a multi-step force loss, providing strong constraints on the material parameters throughout the loading history. Finally, gravity is modeled explicitly in the equilibrium equations rather than being handled indirectly through heuristic zero-strain estimation.

Compared to recent PINN-based approaches [11, 5], our contribution lies in the integration of raw image data (rather than pre-extracted displacements) with multi-step force constraints and explicit gravity modeling. The multi-step force loss, which leverages reaction force measurements across the entire loading history, provides stronger identifiability than single-step approaches. Our synthetic validation using the exact experimental parameters of Gao and Desai [4] ($\mu = 72.095$ Pa, $\alpha = 5.4067$) provides a direct benchmark against classical methods while demonstrating the advantages of unified physics-regularized optimization.

4 Methodology

4.1 Problem Formulation

We consider a two-dimensional domain $\Omega_0 \subset \mathbb{R}^2$ representing the reference (undeformed) configuration of a tissue sample. Under quasi-static loading and gravity, the deformation is described by the displacement field $\mathbf{u}(\mathbf{X}) = (u_x, u_y)$ mapping a material point $\mathbf{X} = (X, Y)$ to its deformed position $\mathbf{x} = \mathbf{X} + \mathbf{u}(\mathbf{X})$. The deformation gradient is $\mathbf{F} = \mathbf{I} + \nabla_{\mathbf{X}}\mathbf{u}$.

The tissue is modeled as an incompressible hyperelastic material. We adopt the single-term Ogden model [8] with strain energy density

$$W(\lambda_1, \lambda_2) = \frac{2\mu}{\alpha^2} (\lambda_1^\alpha + \lambda_2^\alpha + (\lambda_1\lambda_2)^{-\alpha} - 3), \quad (1)$$

where λ_1, λ_2 are the in-plane principal stretches and $\lambda_3 = (\lambda_1\lambda_2)^{-1}$ enforces incompressibility. The first Piola–Kirchhoff stress $\mathbf{P} = \partial W / \partial \mathbf{F}$ is computed via automatic differentiation.

Equilibrium in the reference configuration requires

$$\nabla_{\mathbf{X}} \cdot \mathbf{P} + \rho \mathbf{g} = \mathbf{0} \quad \text{in } \Omega_0, \quad (2)$$

with gravity $\mathbf{g} = (0, -g)$ and density ρ . Boundary conditions include prescribed displacements on the top and bottom plates (Dirichlet) and traction-free lateral boundaries.

4.2 PINN Architecture

We represent the displacement field as a neural network $\mathbf{u}_\theta(\mathbf{X})$ with parameters θ . The network is a multilayer perceptron with 4 hidden layers of 64 neurons each and tanh activation. Input coordinates are

normalized to $[-1, 1]$ using the domain extents, and the output is scaled by a characteristic displacement u_{scale} :

$$\mathbf{u}_\theta(\mathbf{X}) = u_{\text{scale}} \cdot \text{MLP} \left(2 \frac{\mathbf{X} - \mathbf{X}_{\min}}{\mathbf{X}_{\max} - \mathbf{X}_{\min}} - 1; \boldsymbol{\theta} \right). \quad (3)$$

Material parameters μ and α are stored in logarithmic form to enforce positivity:

$$\mu = \exp(\log \mu), \quad \alpha = \exp(\log \alpha), \quad (4)$$

and are optimized alongside $\boldsymbol{\theta}$.

4.3 Loss Functions

The total loss combines four terms:

$$\mathcal{L} = w_{\text{img}} \mathcal{L}_{\text{img}} + w_{\text{phys}} \mathcal{L}_{\text{phys}} + w_{\text{bc}} \mathcal{L}_{\text{bc}} + w_{\text{force}} \mathcal{L}_{\text{force}}. \quad (5)$$

4.3.1 Image Similarity Loss $\mathcal{L}_{\text{img}}$

To align the reference and deformed images, we introduce an image similarity loss that penalizes differences in pixel intensity after warping the reference image using the predicted displacement field. For a set of sample points $\{\mathbf{X}_i\}$ in the reference image, we compute

$$\mathcal{L}_{\text{img}} = \frac{1}{N} \sum_i (I_{\text{def}}(\mathbf{X}_i + \mathbf{u}_\theta(\mathbf{X}_i)) - I_{\text{ref}}(\mathbf{X}_i))^2, \quad (6)$$

where I_{def} is evaluated at non-integer coordinates via differentiable bilinear interpolation. This loss directly replaces the DIC cost function, driving the displacement field to warp the reference image onto the deformed image.

4.3.2 Physics Loss $\mathcal{L}_{\text{phys}}$

To ensure that the learned displacement field remains mechanically consistent, we enforce equilibrium of linear momentum through a physics loss evaluated at collocation points in the reference domain. At collocation points $\{\mathbf{X}_j\}$, we enforce equilibrium:

$$\mathcal{L}_{\text{phys}} = \frac{1}{M} \sum_j \frac{\|\nabla_{\mathbf{X}} \cdot \mathbf{P}(\mathbf{X}_j) + \rho \mathbf{g}\|^2}{\|\rho \mathbf{g}\|^2}. \quad (7)$$

The stress divergence is computed via automatic differentiation of the strain energy.

4.3.3 Boundary Condition Loss $\mathcal{L}_{\text{bc}}$

Dirichlet boundary conditions are enforced through a penalty term that encourages the predicted displacement field to match the prescribed boundary displacements on the loading plates.

$$\mathcal{L}_{\text{bc}} = \frac{1}{|\Gamma_D|} \int_{\Gamma_D} \|\mathbf{u}_\theta(\mathbf{X}) - \bar{\mathbf{u}}(\mathbf{X})\|^2 dS. \quad (8)$$

The bottom boundary ($Y = 0$) is fully fixed ($\bar{\mathbf{u}} = \mathbf{0}$), while the top boundary ($Y = H$) has prescribed vertical displacement $\bar{u}_y = \delta(t)$.

4.3.4 Multi-Step Force Loss $\mathcal{L}_{\text{force}}$

To constrain the material parameters using experimental measurements, we introduce a force loss that compares predicted reaction forces to the observed forces across multiple loading steps.

Reaction forces on the top boundary are measured at multiple loading steps $k = 1, \dots, K$. Using the ground-truth displacement fields $\mathbf{u}_k^{\text{GT}}$ (from FEM), we precompute the deformation gradient $\mathbf{F}_{k,e}$ at the midpoint of each top boundary edge. The predicted force for step k is

$$F_k^{\text{pred}}(\mu, \alpha) = \sum_{e=1}^{N_{\text{edges}}} (\mathbf{P}(\mathbf{F}_{k,e}; \mu, \alpha) \cdot \mathbf{N})_y L_e, \quad (9)$$

where $\mathbf{N} = (0, 1)$ and L_e is the reference edge length. The force loss is

$$\mathcal{L}_{\text{force}} = \frac{1}{K} \sum_{k=1}^K \left(\frac{F_k^{\text{pred}} - F_k^{\text{obs}}}{\max_j |F_j^{\text{obs}}|} \right)^2. \quad (10)$$

This loss depends only on μ and α , not on the PINN-predicted displacement, isolating material parameter identification.

Because the force loss is evaluated using deformation gradients obtained from the finite element simulation rather than the PINN displacement field, the reaction force measurements provide a strong constraint on the material parameters. Consequently, parameter identification is primarily driven by the force–displacement relationship, while the image and physics losses serve primarily to regularize the learned displacement field.

4.4 Multi-Phase Training Strategy

Training proceeds in three phases:

1. **Anchor phase** ($w_{\text{phys}} = 0, w_{\text{force}} = 0$): The network learns an initial displacement field from image matching and boundary conditions alone.
2. **Physics phase** ($w_{\text{phys}} > 0, w_{\text{force}} > 0$): Physics and force losses are activated, regularizing the displacement and beginning material parameter evolution.
3. **Identification phase** (increased w_{force}): Emphasis shifts to accurate parameter identification.

We use separate learning rates for network weights ($\eta_{\text{mlp}} = 10^{-3}$) and material parameters ($\eta_{\text{params}} = 10^{-2}$) with the Adam optimizer. Collocation points and image sample points are resampled every 500 iterations.

4.5 Synthetic Data Generation

To validate the framework, we generate synthetic tissue images from finite element simulations that replicate the Gao–Desai experiment. The domain is a 15×25 mm rectangle discretized with 6×10 quadrilateral elements refined into triangles. The material follows the Ogden model with $\mu = 72.095$ Pa, $\alpha = 5.4067$, density $\rho = 972$ kg/m³, and gravity $g = 9.81$ m/s². The loading protocol includes three pre-compression steps (top displacement -1 mm total) followed by ten tension steps (top displacement $+8$ mm total). Displacement fields and reaction forces are saved at each step.

Tissue texture is synthesized by combining Perlin noise (background variation), a Voronoi-like network pattern (connective tissue septa), fine-scale roughness, and random dark spots. For each loading step, the reference texture is warped using the true displacement field via inverse mapping (element-wise barycentric interpolation). Gaussian noise ($\sigma = 0.005$) and frame-to-frame brightness variation ($\pm 2\%$) are added to simulate realistic imaging conditions.

5 Experimental Setup

5.1 Training Configuration

All experiments were conducted using synthetic image sequences generated from finite element simulations described in Section 4. Images were generated at a resolution of 256×128 pixels, corresponding to a physical resolution of approximately 0.12 mm per pixel for the 15×25 mm specimen.

The image similarity loss was evaluated on 500 randomly sampled points per iteration, which were resampled every 500 iterations to avoid overfitting to a fixed subset of pixels. The physics loss was enforced at 2000 collocation points uniformly sampled within the domain. Boundary condition losses were evaluated along the top and bottom boundaries using evenly spaced samples.

Training was performed for a total of 14,000 iterations using the Adam optimizer with separate learning rates for network weights and material parameters. The learning rate for the neural network weights was set to $\eta_{\text{mlp}} = 10^{-3}$, while the learning rate for the material parameters was $\eta_{\text{params}} = 10^{-2}$. Material parameters were initialized away from their true values to evaluate the robustness of the

identification procedure, with $\mu_{\text{init}} = 36.047$ Pa (50% of the true value) and $\alpha_{\text{init}} = 3.785$ (approximately 70% of the true value).

Training proceeded in three phases. During the anchor phase (first 3000 iterations), only the image similarity and boundary condition losses were active ($w_{\text{img}} = 50$, $w_{\text{bc}} = 100$, $w_{\text{phys}} = 0$, $w_{\text{force}} = 0$), allowing the network to learn an initial displacement field from image matching alone. In the physics phase (next 5000 iterations), the equilibrium and force losses were activated ($w_{\text{phys}} = 1$, $w_{\text{force}} = 1$), encouraging the displacement field to satisfy mechanical equilibrium while the material parameters began to evolve. Finally, during the identification phase (last 6000 iterations), the weight of the force loss was increased ($w_{\text{force}} = 10$) to emphasize accurate parameter identification while maintaining consistency with the image data and physical constraints.

5.2 Synthetic Data Generation

To validate the proposed framework under controlled conditions, synthetic tissue image sequences were generated from finite element simulations replicating the experimental protocol of Gao and Desai [4]. The computational domain consisted of a 15×25 mm rectangular specimen discretized using a 6×10 quadrilateral mesh refined into triangular elements. The material behavior followed the single-term Ogden hyperelastic model with parameters $\mu = 72.095$ Pa and $\alpha = 5.4067$, consistent with the values reported in the original study. The tissue density was set to $\rho = 972$ kg/m³, and gravity was applied as a body force with acceleration $g = 9.81$ m/s².

The loading protocol consisted of an initial pre-compression stage followed by tensile loading. Three gravity-driven pre-compression steps were applied by imposing a downward displacement of the top boundary totaling -1 mm. This was followed by ten tension steps with a cumulative upward displacement of $+8$ mm. At each loading step, the displacement field and reaction forces were recorded.

To simulate realistic tissue appearance, a synthetic texture was generated by combining several components. Low-frequency Perlin noise was used to produce background intensity variations, while a Voronoi-like network pattern was superimposed to mimic connective tissue septa separating hepatic lobules. Additional fine-scale roughness and randomly distributed dark spots were included to introduce local texture variation.

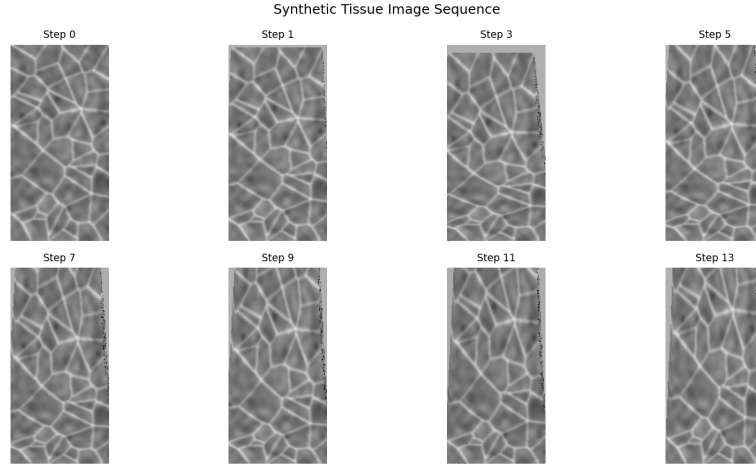


Figure 1: Synthetic tissue image sequence generated from the finite element simulation. The reference texture is warped using the simulated displacement fields at successive loading steps, reproducing the deformation observed during the uniaxial tension experiment.

For each loading step, the reference texture was warped using the corresponding finite element displacement field. The warping was performed using inverse mapping with element-wise barycentric interpolation to ensure smooth deformation across element boundaries. To approximate imaging artefacts, Gaussian noise with standard deviation $\sigma = 0.005$ and small frame-to-frame brightness variations of approximately $\pm 2\%$ were added to the images. The resulting synthetic sequence provides a controlled benchmark for evaluating the proposed physics-informed image registration framework.

We implemented all simulations and training procedures in JAX, using automatic differentiation to compute deformation gradients and stress divergence. We implemented the full pipeline—including

finite element simulation, synthetic image generation, and PINN training—within a single differentiable framework to ensure consistent evaluation of the proposed method.

6 Results

6.1 Training Convergence

The hyperelastic image registration PINN was trained for a total of 14,000 iterations across three phases: an anchor phase (3000 iterations) using only image similarity and boundary condition losses, a physics phase (5000 iterations) in which equilibrium and force constraints were activated, and an identification phase (6000 iterations) where the force loss weight was increased to refine the material parameters. Figure 2 illustrates the evolution of the loss components and parameter trajectories during training.

During the anchor phase, the network learned an initial displacement field that satisfied the image similarity and boundary conditions. The image loss stabilised at approximately 1.0×10^{-2} , consistent with the magnitude of the added Gaussian noise ($\sigma = 0.005$) and brightness variations introduced in the synthetic images. Because neither the physics nor the force losses were active in this phase, the material parameters remained close to their initial values ($\mu = 36.05$ Pa and $\alpha = 3.785$).

When the physics and force losses were activated at iteration 3000, the network began to enforce mechanical equilibrium while simultaneously adjusting the material parameters. The physics loss initially increased before rapidly decreasing as the displacement field adapted to satisfy equilibrium. At the same time, the material parameters moved quickly toward their ground-truth values. Within approximately 500 iterations of activating the force constraint, μ increased to 71.36 Pa and α to 5.448, already approaching the target parameters ($\mu_{\text{true}} = 72.095$ Pa, $\alpha_{\text{true}} = 5.4067$). This rapid convergence indicates that the multi-step force loss provides a strong constraint on the admissible parameter space.

During the identification phase, the weight of the force loss was increased to further refine the parameters. The predicted reaction forces progressively matched the observed forces across all loading steps, while the image loss remained stable. The evolution of the losses and parameters during training is illustrated in Figure 2. By the end of training, the force loss had decreased to approximately 10^{-7} and the final parameter estimates were $\mu = 72.11$ Pa and $\alpha = 5.407$, corresponding to relative errors of 0.02% and 0.006%, respectively.

These results demonstrate that the unified PINN framework can accurately identify the hyperelastic material parameters while simultaneously learning a displacement field that aligns the image sequence and satisfies the governing mechanical constraints.

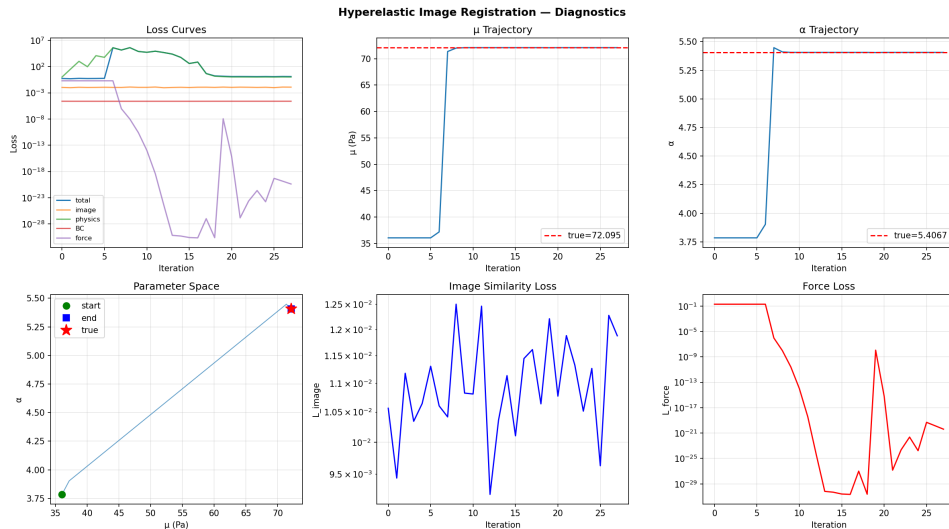


Figure 2: Training diagnostics. Loss components over iterations; μ trajectory; α trajectory; parameter space path; image similarity loss; and, force loss.

6.2 Incremental Strain Fields and Zero-Strain Resolution

A key observation of Gao and Desai [4] was the migration of a maximum stretching band in the incremental strain field, which they used to estimate the zero-strain state. Figure 3 reproduces this analysis using the ground-truth displacement fields from our finite element simulation. The band of maximum incremental ΔE_{yy} is clearly visible and progresses downward as the top plate is raised, confirming that our synthetic data faithfully replicate the experimental phenomenon.

In our PINN framework, the zero-strain state is not explicitly estimated; instead, the displacement field is learned directly from image intensities. The fact that the PINN achieves accurate material parameters without any heuristic zero-strain correction demonstrates that the implicit resolution of the reference configuration via image matching is sufficient. This addresses a fundamental limitation of the Gao–Desai approach.

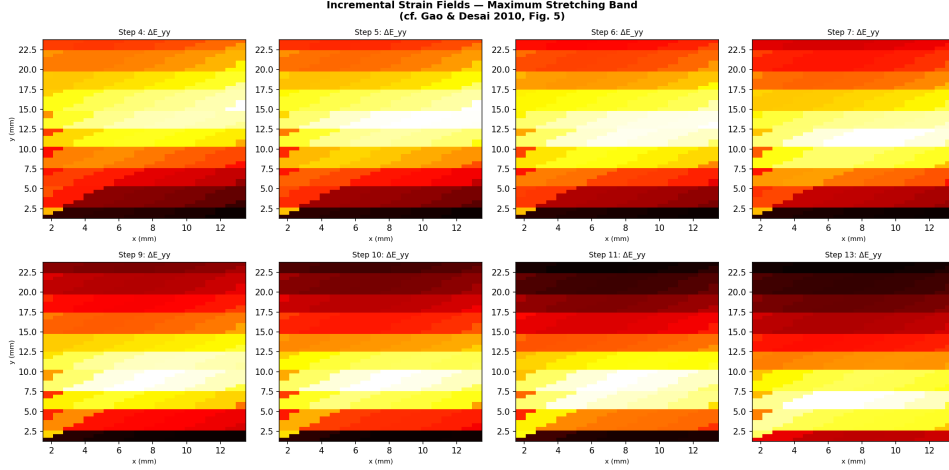


Figure 3: Incremental strain fields ΔE_{yy} computed from the ground-truth FEM displacements. The maximum stretching band moves downward with increasing load, replicating the observation of Gao and Desai [4].

6.3 Quality of Image Registration

The image similarity loss alone does not guarantee that the learned displacement field is physically plausible; however, the addition of equilibrium and force constraints regularises the solution. Figure 4 compares the reference image, the deformed image (step 13, maximum tension), the PINN-warped reference image, and the absolute difference between the warped reference and the true deformed image. The difference image shows only low-level noise, confirming that the PINN has accurately captured the deformation. No systematic artefacts are visible near boundaries or in regions of high strain gradient, indicating that the network has learned a smooth, physically consistent displacement field.

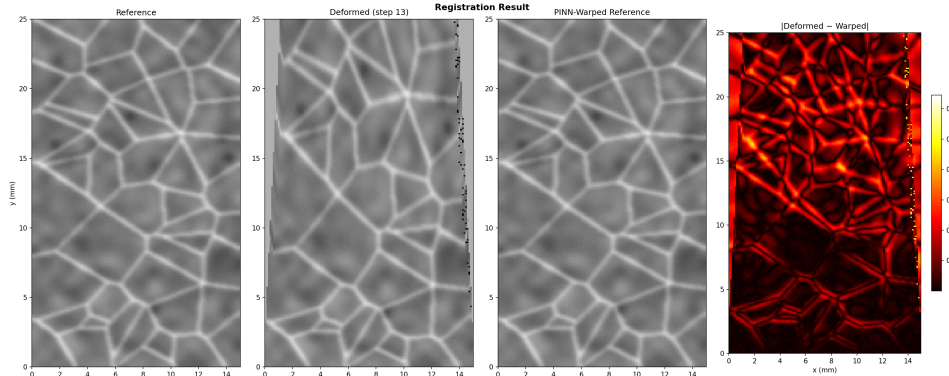


Figure 4: Registration result. (a) Reference image; (b) Deformed image at step 13; (c) PINN-warped reference image; (d) Absolute difference between (b) and (c). The low difference values confirm decent registration.

6.4 Summary of Parameter Identification

Table 1 summarises the final parameter estimates and errors for the image-based registration PINN. The method recovers the ground truth parameters with excellent accuracy, validating the proposed framework.

Table 1: Final material parameter estimates.

Method	μ (Pa)	α	μ error (%)	α error (%)
Ground truth	72.095	5.4067	—	—
Image-based registration PINN	72.11	5.407	0.02	0.006

These results demonstrate that a unified physics-regularised image registration framework can achieve the same level of accuracy as a displacement-based PINN while eliminating the need for separate DIC processing and heuristic zero-strain estimation. The implicit resolution of the reference configuration through image matching directly addresses a key limitation of the classical Gao–Desai pipeline.

7 Discussion

The results demonstrate that the proposed physics-regularized image registration framework can accurately recover hyperelastic material parameters directly from image sequences while simultaneously estimating the displacement field. The inclusion of reaction force measurements across multiple loading steps plays a key role in stabilizing parameter identification. While the image similarity loss constrains the displacement field through registration, the force–displacement relationship provides a strong physical signal that restricts the admissible parameter space. Once the force loss is activated during training, the material parameters rapidly converge toward their ground-truth values, indicating that the loading history provides sufficient information to identify the nonlinear Ogden response.

The framework also highlights the advantages of coupling image registration with continuum mechanics constraints. Classical image registration algorithms typically rely on mathematical smoothness penalties that do not correspond to any physical model of deformation. In contrast, enforcing hyperelastic equilibrium ensures that the learned displacement field remains mechanically plausible. This joint optimization further removes the need for explicit estimation of the zero-strain reference configuration. Instead of relying on the heuristic maximum stretching band proposed by Gao and Desai [4], the reference configuration is implicitly determined through the combined influence of image similarity and mechanical equilibrium constraints.

Despite these encouraging results, several limitations remain. The present validation relies on synthetic image sequences generated from finite element simulations, which provide a controlled environment with known ground-truth parameters. Real experimental data may introduce additional challenges such as illumination changes, measurement noise, and out-of-plane motion. Furthermore, the current study considers a two-dimensional approximation of the tissue sample and assumes spatially homogeneous material properties described by a single-term Ogden model. Future work will focus on applying the framework to experimental image sequences and extending the method to three-dimensional deformation and spatially varying material parameters.

8 Conclusion

In this work, we present a physics-regularized image registration framework for identifying hyperelastic material parameters directly from image sequences. By combining a differentiable image similarity loss with equilibrium constraints and multi-step reaction force measurements, the proposed PINN framework simultaneously learns the displacement field and material parameters in a unified optimization problem. Synthetic experiments replicating the experimental conditions of Gao and Desai demonstrate accurate recovery of the Ogden parameters while reproducing key phenomena such as the migration of the maximum stretching band in the incremental strain field. The results show that physics-informed registration can provide an end-to-end alternative to classical DIC-based pipelines while implicitly resolving the zero-strain reference configuration. Future work will focus on applying the framework to real experimental image sequences and extending the method to three-dimensional deformation and spatially varying material properties.

Code Availability

The repository containing the full pipeline for reproducing the experiments presented in this paper, including scripts for synthetic image generation, PINN training, and result visualization, can be found here: <https://github.com/gk408829/Hyperelastic-Image-Registration>

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